

The emissivity of a body depends on both the temperature and the nature of emitting surface. Bodies of same temperature and size but different material emit different amount of total thermal radiation. According to Kirchoff's Law good absorbers are good emitters. For Theoretical purpose we define an ideal substance capable either of absorbing all the thermal radiation falling on it or emitting all the energy provided to it in the form of thermal radiation. Such a substance is called Black Body.

Thus for a black body we can say that its emissivity is 1, as well as its absorptivity. We can compare different substances with black body and obtain the emissivity of respective substances. For example a polished shining steel piece emits only 9% of the thermal radiation as compared to a black body of same size and shape of same temperature, hence its emissivity is 0.09. Similarly emissivity of a rough oxidized steel surface is 0.81 and that of ocean water is 0.96 and some substances like lamp black, coal are there whose emissivity is very nearly unity that is, almost an ideal emitter. The emissivity of a general body is denoted by e and mathematically defined as

$$e = \frac{\text{Radiant power emitted by body at a given temperature}}{\text{Radiant power emitted by a Black Body of same geometry at same temperature}}$$

Thus total thermal radiation power emitted by a General Body is given as

$$\left(\begin{array}{c} \text{Radiation Power} \\ \text{from a} \\ \text{General Body} \\ \text{at a given temperature} \end{array} \right) = e \left(\begin{array}{c} \text{Radiation Power} \\ \text{from a} \\ \text{Black Body} \\ \text{at same temperature} \end{array} \right) \quad \dots (4.30)$$

Experimentally a very good approximation to a black body is provided by a cavity enclosed by high temperature opaque walls regardless of the composition of the material of its interior walls.

4.3.3 Stefan Boltzmann Law

On the basis of experiments, Stefan concluded in 1879 that the total amount of thermal radiation power from a body is directly proportional to the fourth power of its absolute temperature. The mathematical result was derived later by Boltzmann that the total radiation power per unit surface area of a body is given as

$$P_R = \sigma T^4 \text{ J/s.m}^2 \quad \dots (4.31)$$

If a body of surface area A is at temperature T , then total thermal power radiated by the body is

$$P_R = \sigma A T^4 \text{ J/s} \quad \dots (4.32)$$

Later experimentally it is observed that equation-(4.32) does not give correct results in many cases, it gives approximately true results for substances with high emissivities and it was accepted that equation-(4.32) is only valid for black bodies or those having emissivity very close to unity. Thus for general

bodies equation-(4.32) can be generalized as power radiated by a general body of surface area A , at temperature T and with emissivity e , can be given by equation-(4.30) as

$$P_R = e A \sigma T^4 \text{ J/s} \quad \dots (4.33)$$

Any object not only emits energy by radiation, but it also absorbs energy radiated by other bodies in its surrounding. If an object of emissivity e and area A is at a temperature T , it radiates energy at a rate $e A \sigma T^4$. If the object is surrounded by an environment at temperature T_s . The rate at which the surrounding in immediate contact of the body surface radiate energy at a rate $\sigma A T_s^4$ if it is assumed to be a black body and if absorptivity of the body material is a , the amount of energy absorbed by the body is at a rate $a \sigma A T_s^4$. Thus the net heat flow from the object can be given as

$$\frac{dQ}{dt} = e A \sigma T^4 - a A \sigma T_s^4 \quad \dots (4.34)$$

As good absorbers are good emitters we can approximately use $e \approx a$ thus we have

$$\frac{dQ}{dt} = e A \sigma (T^4 - T_s^4) \quad \dots (4.35)$$

When body and surrounding both are at same temperature equation-(4.35) shows that there is no net flow of heat takes place with the body. When $T > T_s$, net flow of heat takes place from body to surrounding and temperature of body decreases. Similarly if $T < T_s$, net flow of heat takes place from surrounding to body and temperature of body increases with time.

If mass of body is m and s is the specific heat of the material of body then we have

$$\frac{dQ}{dt} = ms \frac{dT}{dt} \quad \dots (4.36)$$

Where $\frac{dT}{dt}$ is rate of change of temperature of body. If $\frac{dQ}{dt}$ is rate of heat loss from a body than $\frac{dT}{dt}$ will be the rate at which temperature of body falls and is termed as rate of cooling. Thus if a body of mass m and specific heat s at temperature T is placed in surrounding of temperature T_s then the rate of cooling of body can be given as

$$\frac{dT}{dt} = \frac{e A \sigma}{ms} (T^4 - T_s^4) \text{ K/s} \quad \dots (4.37)$$

4.3.4 Solar Constant

It is defined as "solar energy received per unit surface area of earth surface per unit time, when sun is overhead". If radius of sun is R_s and its temperature T_s , the average rate at which it radiates energy is given as

$$E = \sigma (4\pi R_s^2) T_s^4 \quad \dots (4.38)$$